

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Completing the Square

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

COMPLETING THE SQUARE · CH2 · L5



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The Map

This lesson collapses Completing the Square into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Halve the coefficient of x

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Add and subtract the square

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Find turning point form

TRIGGER *"work out", "find", a missing value*

04 Solve using completed square

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Evaluate fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 29 website questions, 29 checked solutions, 3 local PDFs read.

01 Halve the coefficient of x

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Completing the Square question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q23

Solve inequality

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

Finish: $25 - 2(x + 3)^2$

FORMULA BANK

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \left(\frac{b}{2}\right)^2.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Add and subtract the square

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Completing the Square question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
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WORKED MODEL

Source pattern: Jan 2020 P2HR Q23

Solve inequality

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

Finish: $25 - 2(x + 3)^2$

FORMULA BANK

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \left(\frac{b}{2}\right)^2.$$

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

23 Express $7 - 12x - 2x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

YOUR SOLUTION

03 Find turning point form

TRIGGER *"work out", "find", a missing value*

BECOMES A Completing the Square question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$28 + 24x - 6x^2 = -6x^2 + 24x + 28$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2025 P2H Q25

(a)

$$28 + 24x - 6x^2 = -6x^2 + 24x + 28$$

(b)

$$y = 52 - 6(x - 2)^2$$

Finish: (a) $52 - 6(x - 2)^2$. For the sketch, show a downward parabola with turning point (2, 52) and y -intercept (0, 28).

FORMULA BANK

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \left(\frac{b}{2}\right)^2.$$

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

25 (a) Write $28x^2 - 6x + 2$ in the form $a(x - c)^2 + d$ where a , b and c are integers. (3) (b) On the axes below, sketch the graph of $y = 28x^2 - 6x + 2$. Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the y -axis. (3)

YOUR SOLUTION

04 Solve using completed square

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Completing the Square question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q23

Solve inequality

$$7 - 12x - 2x^2 = -2x^2 - 12x + 7$$

Finish: $25 - 2(x + 3)^2$

FORMULA BANK

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \left(\frac{b}{2}\right)^2.$$

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

23 Express $7 - 12x - 2x^2$ in the form $a + b(x + c)^2$ where a , b and c are integers.

YOUR SOLUTION

05 Evaluate fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Completing the Square question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\sqrt[3]{\frac{1}{x}} = x^{-1/3}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1HR Q19

(a)

$$\sqrt[3]{\frac{1}{x}} = x^{-1/3}$$

Evaluate fraction

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $m = -\frac{1}{3}$, and $3(x+2)^2 + 7$

FORMULA BANK

$$x^2 + bx + c = (x + \frac{b}{2})^2 + c - (\frac{b}{2})^2.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

19 Given that $13 \times x^m$ (a) find the value of m $m = (1)$ Given that a, b and c are integers, (b) express $3x^2 + 12x + 19$ in the form $a(x + b)^2 + c$ (2)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Halve the coefficient of x
"work out", "show", "hence", a familiar exam phrase	Add and subtract the square
"work out", "find", a missing value	Find turning point form
"work out", "show", "hence", a familiar exam phrase	Solve using completed square
"work out", "show", "hence", a familiar exam phrase	Evaluate fraction
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.